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Incidence of pain during cesarean delivery with neuraxial anesthesia: an international, prospective cohort study

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Abbreviations

CI Confidence Interval

CSE Combined Spinal and epidural

DPE Dural puncture epidural

IRB institutional review board

IQR interquartile Range

SD Standard Deviation

SOAP Society for Obstetric Anesthesia and Perinatology

STROBE Strengthening the Reporting of Observational Studies in Epidemiology

Abstract

Background: Neuraxial anesthesia techniques are considered the preferred technique for cesarean delivery, however the true incidence of pain during cesarean delivery with neuraxial anesthesia is not currently known. To date studies have been retrospective or conducted in a single center. The primary aim of this international cohort study was to prospectively determine the incidence of patient reported pain during cesarean delivery with neuraxial anesthesia.

Methods: We conducted, over 8 weeks, a multicenter cohort study in 15 centers across the United States of America and Canada; all patients who underwent cesarean delivery with neuraxial anesthesia were surveyed on postpartum day 1 regarding the presence of intraoperative pain, grade their pain using a numeric rating score (0-10) and rate their satisfaction (Yes/No) with their pain management..

Results: A total of 3,693 patients were included in the analysis. An overall incidence of patient reported pain during cesarean delivery was 7.6% (95% CI 6.8%, 8.5%) with 282 of 3,693 patients reporting pain. In patients undergoing elective cesarean delivery, intraoperative pain was reported in 3.7% (95% CI 2.7%, 5.0%) with spinal, 9.2% (95% CI 6.7%, 12.2%) with combined spinal-epidural and 12.2% (95% CI 4.1%, 26.2%) with epidural top-up anesthesia. In the non-elective setting, pain was more commonly reported by patients, 5.7% (95% CI 4.0, 7.8) with spinal, 7.1% (95% CI 4.7%, 10.1%) with combined spinal-epidural, 8.0% (95% CI 2.2%, 19.2%) with dural puncture epidural and 13.2% (CI 95% CI 11.1%, 15.5%) with epidural top-up anesthesia. Patients who had intraoperative pain reported a median (IQR) numeric rating score of 6 (4-8) out of 10. In those who reported pain, dissatisfaction with how the pain was managed by the anesthesia team was reported by 29 (10.3%) patients.

Conclusions: Overall incidence of patient reported pain during cesarean delivery with neuraxial anesthesia in this multicenter cohort was 7.6%. This varied significantly by anesthesia technique, with spinal having the lowest incidence and epidural top-up the highest. Further work is required to characterize the pain experience, understand its impact, and develop techniques to reduce its incidence.

INTRODUCTION

Cesarean delivery rates continue increase worldwide, with over 30 million performed each year. This is predicted to increase to nearly 30% of all deliveries by 2030.[1] Neuraxial anesthesia techniques are the preferred technique for cesarean delivery, with its use associated with superior outcomes and reduced morbidity for both mother and baby.[2–5] However, adequate neuraxial anesthesia is not completely guaranteed and previously reported incidences of “inadequate anesthesia” for cesarean delivery have ranged from 0.8 to 15%.[4,6–8]

Previous studies have reported marked variation in how intraoperative pain has been defined and approached.[4,7,9–17] Many studies have been single center, retrospective and relied on surrogates of patient reported intraoperative pain, for example conversion to general anesthesia, repeat neuraxial procedure, or unplanned supplemental analgesia or sedation.[6,7,9,18,19] However, not all patients who report intraoperative pain receive supplemental interventions.[20] Further, proxies may be affected by investigator assessments and clinical practices.[21,22] The American Society of Anesthesiologists have highlighted a need to focus on patient reported pain to avoid the risk of significantly underestimating the true frequency of pain during cesarean delivery.[23]

Analysis of anesthesia related litigation claims in the United Kingdom found a higher incidence of cases due to inadequate neuraxial anesthesia compared to claims from general anesthesia, with the majority occurring in the obstetric setting.[24,25] There can be significant adverse sequelae from the experience of pain during cesarean delivery. Inadequate neuraxial anesthesia can lead to psychological morbidity in the short and longer term, including an increased incidence of postpartum depression[26] and post-traumatic stress disorder.[27,28]

The primary aim of this study was to prospectively determine the incidence of pain during cesarean delivery across multiple North American centers. The secondary aims were to explore demographic, obstetric, anesthetic associations with intraoperative pain, identify institutional differences in intraoperative pain and assess satisfaction of intraoperative pain management.

MATERIALS AND METHODS

Study design

We conducted a prospective, multicenter cohort study across participating centers in the United States of America and Canada. We invited centers to participate via the Society for Obstetric Anesthesia and Perinatology (SOAP) research network. This study was conducted and is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies [29]. The study was registered with the SOAP research network and the protocol was published on open source prior to data analysis. The data analysis and statistical plan were written and filed with Stanford University institutional review board (IRB) protocol number 70611, 2nd August 2023 before data were accessed; each center gained local IRB approval and data sharing agreements with Stanford University. Following approval by each center's local IRB, the need for informed consent was waived either by designation of the data collection as a quality improvement initiative exempted from IRB review or by explicit waiver of informed consent for research by the local IRB. The participating center agreed to collect data for a minimum of 8 weeks consecutively, with a goal of enrolling at least 95% of patients who underwent cesarean delivery with a neuraxial anesthetic during the data collection period.

Study population

As informed consent was not required, all patients who delivered via cesarean delivery under a neuraxial anesthetic during the study period were eligible. We included patients of all ages and any viable gestational age. We excluded patients who refused, were unable to participate or answer the questions, for example, due to admission to intensive care and those who were anesthetized with a general anesthesia prior to surgical incision.

For at least 8 weeks at each center, patients were approached and interviewed 24 (+/- 8) hours postpartum between August 2023 and June 2024. Each participating center attended two study group meetings, and a further virtual site initiation visit to instruct the conduct of the study. To limit intra- and inter- institutional variability and reduce potential bias, local study investigators, distinct from the direct care team, used a predefined script (Supplementary information, <https://links.lww.com/ALN/E299>) where patients were asked the binary question whether they experienced pain during their surgery (Yes/ No). If they reported experiencing pain, they were asked to grade the severity using a numeric rating score (0-10; 0=no pain to 10=worst pain imaginable) and whether they were satisfied (Yes/No) with the intraoperative management of the pain by the anesthesiology team. In addition, patient demographic (age, language, race and ethnicity), anesthetic (mode of neuraxial and requirement for general anesthesia and critical care admission), and obstetric (urgency of cesarean, categorized as elective, urgent (“maternal or foetal compromise that is not immediately life threatening”) or emergency (“Immediate threat to life of mother or foetus, goal to deliver within 30 minutes”) and intrapartum status data were collected from the patient and their electronic health record. The data were then transcribed to a deidentified data collection sheet Excel v. 16.6 (Microsoft, Inc., Redmond, WA, USA.) before being sent securely via data link to Stanford University for analysis. Patients who reported

intraoperative pain were invited to further characterize their pain experience as part of a subsequent consenting study.

Statistical methods

The total sample size would depend on the total number of recruiting centers contributing data. We estimated that on average each center would enroll approximately 200-300 patients and have a 15% incidence of intraoperative pain.[9] We therefore aimed to have a minimum sample size of 545 patients which would be sufficient to provide a 95% confidence and 3% margin of error. Data were collated using Excel v. 16.6 (Microsoft, Inc., Redmond, WA, USA.) then subsequently cleaned with the correction of corrupted, incorrectly formatted or incomplete data and statistical analyses performed Stata v. 18.0 (Statacorp, College Station, TX, USA) and SAS 9.4 (SAS Institute Inc., Cary, NC, USA).

Descriptive analyses were conducted to describe patient demographic, obstetric and anesthetic-related factors of those with intraoperative pain. Shapiro-Wilk test was used to assess for normality of continuous variables. Mean (SD) or median (IQR) were calculated for normally and non-normally distributed continuous variables, respectively. Categorical variables were summarized as frequencies and percentages. An uncorrected Chi-square test or Fishers' exact test were used to compare the frequencies of categorical variables, and one-way analysis of variance or Kruskal-Wallis test used to compare continuous variables between groups. We calculated unadjusted and adjusted relative odds ratios for each measured variable. We performed multilevel logistic regression modelling using the GLIMMIX procedure with maximum likelihood estimation based on the Laplace approximation. We conducted two models, a null model including the hospital as a random effect and a second model with the addition of patient factors to determine the extent of the hospital-level variation. In each model, we quantified the between-hospital variation using

random-effects variance by computing the median odds ratio (MOR) and the intraclass coefficient (ICC). Further, we assessed for multicollinearity of measured variables using variance Inflation Factor (VIF) The aggregated data are available upon reasonable request and IRB approval.

RESULTS

A total of 3,758 patients underwent a cesarean delivery during the recruitment period in the 15 participating centers in the United States of America and Canada. A list of participating institutions is found in appendix 1. Figure 1. shows the STROBE study flow chart of patients for the study with 3,693 patients being included in the analyses. A total of 1,684 (45.6%) of these were elective and 2009 (54.4%) were non-elective cesarean deliveries. Of the included patients, 1,810 (49.0%) had spinal anaesthesia, 855 (23.2%) combined spinal-epidural (CSE), 53 (1.4%) DPE and 975 (26.4%) epidural top-up for their cesarean delivery. The baseline demographic, obstetric and anesthetic data of those with and without pain are shown in table 1.

An overall incidence of patient reported pain during cesarean delivery was 7.6% (95% CI 6.8%, 8.5%) with 282 of 3,693 patients reporting pain. Pain was reported by 80 of 1,810 (4.4% (95% CI 3.5% 5.5%)) patients with spinal anesthesia. In those who underwent epidural top-up, 13.1% (95% CI 11.1%, 15.4%) or 128 of 975 of patients reported pain. A total of 70 of 785 patients (8.9% (95% CI 6.4%, 10.2%)) who had CSE and 4 of 49 (7.6% (95% CI 2.1%, 18.2%)) of patients who had dural puncture epidural anesthesia had pain reported during their cesarean delivery. Pain was overall more commonly reported in the non-elective setting 9.5% (95%CI 8.2,10.8) vs 5.4% (95% CI 4.3, 6.5), $p=0.04$. Table 2 reports the overall incidence and pain scores for each anesthesia mode in the elective and non-elective settings. In unadjusted models, Spanish as a first language, CSE and epidural top-up anesthesia for cesarean delivery were

associated with higher odds of reporting intraoperative pain, 1.98 (95% CI 1.33, 2.95), 1.93 (95% CI 1.38, 2.68) and 3.27 (95% CI 2.44, 4.37), respectively.

Table 3 highlights adjusted odds ratios for the measured variables. There were no associations observed in categorization of cesarean urgency or intrapartum status with patient reported intraoperative pain. The VIFs of measured variables were less than 10, indicating a low to moderate level of multicollinearity. The MOR for the unadjusted and adjusted models were 1.49 (95% CI 1.25, 1.97) and 1.47 (95% CI 1.21, 1.97), respectively. Therefore, a patient's odds of intraoperative pain significantly varied by hospital and a patient in a hospital with higher baseline risk had approximately 47% high median odds of pain compared to a similar patient in a hospital with lower baseline risk, when controlling individual patient level factors. The ICC for unadjusted and adjusted models were 5.1 (95% CI 1.7, 13.4) and 4.7 (95% CI 1.2, 13.4) respectively. The MOR and ICC did not alter significantly in adjusted models, meaning hospital level variation was not explained by patient level factors.

The median (IQR) numeric rating score for all patients with pain was 6 (4-8) out 10, with statistically higher pain scores in those undergoing non-elective compared to elective cesarean deliveries, 7 (4-9) vs 6 (4-8) respectively. There were no significant differences in pain scores when comparing caesarean urgency and mode of anesthesia. Further, there were no differences between institution, ethnicity, preferred language, or intrapartum status in pain scores.

Regarding patient satisfaction, in patients who reported pain, 29 (10.3%) were dissatisfied with how their pain was managed by the anesthesia team. Those who reported satisfaction in how their pain was managed were less likely to have had intrapartum cesarean delivery, OR (95%CI) 0.41 (0.18, 0.97), a dural puncture epidural 0.02 (0.002, 0.4), have French as their preferred language 0.21 (0.04, 0.99) or be of Black racial background, 0.32 (0.12-0.87) in unadjusted models. Those

who were dissatisfied reported higher pain scores than those who were satisfied, 8 (8-10) vs 6 (4-8) $p=0.0001$. A total of 30 (10.6%) patients who had reported pain had conversion of neuraxial to general anesthesia.

Intraoperative pain during cesarean delivery varied by institution. The lowest overall crude incidence was 3.5% (95%CI 1.8%, 6.1%) and the highest was 11.3% (95%CI 8.4%, 14.8%). In those having spinal anesthesia, the lowest reported incidence of pain was 0 (95%CI 0, 12.8%) and the highest was 11.9% (95%CI 5.3%, 22.2%). The distribution across centers for all modes and spinal anesthesia are found in Figure 2a-b and 3, respectively. There were no differences in the overall incidence of pain between institutions in Canada or the US, mean (95% CI) 7.5% (6.5%, 8.5%) and 8.0% (6.4%, 9.7%), respectively.

DISCUSSION

Our study revealed an overall incidence of 7.6% for patient reported intraoperative pain during cesarean delivery conducted with neuraxial anesthesia. Spinal anesthesia had the lowest incidence of 4.4% or approximately 1 in 23, and epidural top-up had the highest of 13.1% or approximately 1 in 7 patients. When correcting for the measured variables, combined spinal-epidural, epidural anesthesia and preferred language of Spanish were associated with greater risk of pain. The MOR of the adjusted model was 1.47, means a patient's odds of intraoperative pain significantly varied by hospital. Patient dissatisfaction with pain management by their anesthesiology team was associated with the reporting of higher pain scores. Conversion to general anesthesia occurred at a rate of 10.6% in patients who reported intraoperative pain. A small number of patients received de novo epidural anesthesia in the elective setting, representing approximately 1% of cases, which were conducted where clinically indicated.

Clinical relevance

Our study provides contemporary estimates of the incidences of patient reported pain with each neuraxial anesthetic technique, derived from multiple institutions across North America. These can be used to guide clinical context and consent processes. The frequency of pain occurring during cesarean delivery highlights the importance of consent and appropriate communication with patients about the potential risk of intraoperative pain. Clinicians should also consider incorporating shared decision-making for different anesthesia techniques and the management of pain if it occurs.

When correcting for hospital center, age, ethnicity, race, language, urgency and intrapartum status, we found significant differences in unadjusted and adjusted models for the risk of intraoperative pain related to anesthesia mode. Patients who had an epidural top-up were approximately 3 times greater risk of reporting intraoperative pain compared to those who had cesarean delivery performed with spinal anesthesia. This has implications for clinical practice and may heighten clinician awareness regarding identifying and managing intraoperative pain when cesarean delivery is performed using an epidural top-up technique. For all neuraxial techniques, appropriate testing and ensuring sufficient upper and lower sensory block levels should be prioritized. It may also lower the threshold for abandonment as the primary technique if adequate anesthesia is not established prior to cesarean delivery.

The pain reported during cesarean delivery in this study is of high severity. In those who had intraoperative pain, the median numeric rating score for pain would be considered at least “moderate” and those undergoing urgent, or emergency cesarean delivery considered “severe”[30]. Pain during cesarean delivery is consider a significant concern for patients, with

49% ranking pain as the least desirable anesthetic outcome related to their cesarean delivery, before postoperative pain, vomiting, nausea or itching[31].

Caution should be taken with the potential impact of language used by clinicians, including suggesting ‘discomfort’ rather than ‘pain’, and to avoid the ‘nocebo effect’,[32,33] Further, when a patient experiences pain, it should be acted upon appropriately[34,35]. We used scripted, open questions regarding the presence of pain and satisfaction on day 1 postpartum to ensure we maintained patient centrality to not qualifying their experiences[36].

In the US, analysis of 10 years of closed malpractice cases in obstetric anesthesia, representing 3.7% of all specialty cases found 18 associated with major or minor maternal injury and neuraxial anesthesia[37]. Most patients received neuraxial anesthesia, with only 8 patients (7.5%) having general anesthesia. Three of these (17%) related to “inadequate block causing pain during cesarean delivery”. Analysis of litigation claims in the United Kingdom found a higher incidence of cases due to neuraxial anesthesia failure compared to claims from general anesthesia, with the majority occurring in the obstetric setting. A large proportion of these claims related to pain during cesarean delivery and failed or repeated attempts at neuraxial blockade[24,25].

Patients should be appropriately counselled and consented preoperatively, this discussion should incorporate the management of intraoperative pain during cesarean delivery and possible requirement for general anesthesia. Our results can aid providing additional information to ensure the patient is fully informed.

Research context

A previous retrospective single center audit conducted in the United Kingdom 20 years ago evaluated 4540 cesarean deliveries performed with neuraxial anesthesia and found the overall incidence of physician documented pain with spinal, CSE and epidural top-up anesthesia to be

3.5%, 13.4% and 13.7%, respectively [36]. Whereas a recent systematic review and meta-analysis found the overall incidence to be approximately 17% (95%CI 13-22%), with patients who received spinal anesthesia reporting the lowest and those who received epidural top-up the highest incidence of pain of 14% (95%CI: 10-20%) and 33% (95%CI: 17%- 54%), respectively [38].

We asked patients objectively whether they had pain intraoperatively and to quantify their pain. Pain is defined as “An unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage [37]”. In many studies a proxy for patient reported pain are used including the requirement for supplemental analgesia/sedation or general anesthesia. However, not all those with pain received analgesia and not all receiving analgesia report pain, this could represent an underestimate or overestimate of the true incidence of patient reported pain experienced [20]. The incidence of conversion to general anesthesia were low in those who reported pain, highlighting this as a limited surrogate measure of intraoperative pain.

Those who stated their preferred language to be Spanish were almost twice as likely to report intraoperative pain in adjusted models. However, we did not see other disparities related to other recorded languages, racial or ethnic groups of our measured variables in adjusted models. Differences in patient reported pain may be influenced by barriers in language and communication challenges. These barriers can lead to misapprehensions between patients and healthcare providers regarding the presence and severity of pain. Cultural differences and the use of medical jargon can further complicate communication, resulting in inadequate pain management and intervention. There are known disparities related to racial ethnic background with access, experience and outcome in obstetric anesthesia. Asian or Black patients are less likely to have labor neuraxial analgesia and those from Black racial ethnic backgrounds are more likely to receive general anesthesia for cesarean delivery[39–41].

Further work is required to understand the patient, anesthetic, obstetric and institutional level risk factors for intraoperative pain. There are known institutional and regional variations in patient demographics, obstetric care and outcomes, the impact of this related to pain is yet to be fully elucidated[42–45]. In those who are unfortunate to have intraoperative pain, characterization of the pain experienced from the patients’ perspective is necessary. Further, investigation into the sequelae of pain during cesarean in the short- and medium-term postpartum are warranted.

Strengths and limitations

This study has a number of strengths. It is the largest prospectively collected cohort of data on patient reported pain during cesarean delivery. It encompasses a diverse cohort of patients across 15 centers in two countries with differing health system structures. As a quality improvement “capture all” study we were able to include more than 95% eligible patients in our final analysis and included all patients who delivered with a neuraxial technique as the primary mode of anesthesia during the study period, therefore extending the generalizability of this study. In addition, this study addresses the limitations of previous studies, specifically including multiple centers, all categories of cesarean delivery urgency and provides estimates for the incidence with different anesthetic techniques. There are limitations to our study. We invited centers to collaborate via the SOAP research network and to SOAP members, not all centers took part, and the participating centers were self-selected. The centers partaking were academic medical centers, with the majority having SOAP Center of Excellence designation[46], therefore they may not fully represent practice elsewhere. Further, clinicians were not blinded to the data that were being collected, and this may have informed changes to routine practice during the study period. There is no perfect time to interview patients, we asked patients to report pain one day after surgery in line with their routine follow up and therefore aligns with anesthesiology team clinical practice.

This however relies on the patient recalling details of the cesarean delivery and may be influenced by presence or absence of pain between 12-36 hours postpartum. Not all those who had pain may have recalled its presence and or severity. However, this reflects standard timing for anesthesia follow up in most institutions and mirrors experiences with long-term psychological salience. To ensure patient anonymity, we did not collect data of the time or date of delivery, which limited further analysis of temporal associations related to research publications, weekend or time of day. We did not assess association between conversion to general anesthesia and patient satisfaction as the total numbers were small and beyond the scope of the study. All patients were enrolled without the need for written consent, this maximized the inclusion, however this limited the granularity of data we were able to collect for example drug dosing, surgical duration or surgical complexity.

CONCLUSION

In this multicenter study, we found the overall incidence of intraoperative pain during cesarean delivery with neuraxial anesthesia to be 7.6%, as reported by patients in the early postpartum period. The incidence was lowest with spinal anesthesia and highest (13.6%) with epidural anesthesia. These findings may underestimate actual intraoperative pain but reflect experiences most likely to have lasting psychological impact. This has consequences for the perioperative cesarean consent process and selection of neuraxial anesthesia. Best practice preoperative discussions acknowledging the potential for intraoperative pain and relative probabilities of risks should be discussed with patients, whilst considering an understanding of patient preferences. Further work is required to prospectively characterize intraoperative pain experience from the patients' perspective, identify how to prevent pain during caesarean delivery and identify best practices for its management to improve patient satisfaction when it does occur. In addition, the

sequelae of pain during cesarean in the short- and medium-term postpartum needs further investigation.

Supplemental Digital Content

Multicenter Intraoperative Discomfort during Cesarean Delivery study MID-CD,

<https://links.lww.com/ALN/E299>

Accepted Preproof

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Figure Legend

Figure 1. STROBE flow diagram of the study population.

Figure 2a-b. Crude and adjusted incidence of intraoperative pain for all anesthesia modes across center

Figure 3. Crude incidence of intraoperative pain with spinal anesthesia across center

Table 1. Patient demographic, anesthetic and obstetric variables

Table 2. Incidence of intraoperative pain and numerical pain scores by mode of anesthesia in the elective and non-elective cesarean deliveries.

Table 3. Adjusted odds ratios of measured variables for intraoperative pain.

Table 1. Patient demographic, anesthetic and obstetric variables

	Pain reported n=282	No pain reported n=3411	p value
Age years Mean (SD)	32.8±5.8	33.4±5.5	0.065
Ethnicity			0.116
Hispanic n (%)	55 (19.5%)	515 (15.1%)	
Non-Hispanic n (%)	178 (63.1%)	2325 (68.2%)	
Missing n (%)	49 (17.4%)	571 (16.7%)	
Race			0.184
White n (%)	97 (34.4%)	1323 (38.8%)	
Asian n (%)	45 (16.0%)	629 (18.4%)	
Black or African American n (%)	41 (14.5%)	458 (13.4%)	
Indian or Alaska native n (%)	0	17 (0.5%)	
Native Hawaiian or other Pacific Islander n (%)	0	10 (0.3%)	
Mixed Race n (%)	1 (0.4%)	0	
Other Race n (%)	50 (17.7%)	442 (13.0%)	
Missing n (%)	48 (17.0%)	532 (15.6%)	
Preferred Language			0.004
English n (%)	184 (65.3%)	2448 (71.8%)	
Spanish n (%)	33 (11.7%)	221 (6.5%)	
French n (%)	10 (3.6%)	101 (3.0%)	
Other n (%)	6 (2.1%)	132 (3.9%)	
Missing n (%)	49 (17.4%)	509 (14.9%)	
Cesarean urgency			<0.001
Elective n (%)	91 (32.3%)	1593 (46.7%)	
Urgent n (%)	142 (50.4%)	1368 (40.1%)	
Emergency n (%)	49 (17.4%)	450 (13.2%)	
Intrapartum cesarean			<0.001
Intrapartum n (%)	155 (55.0%)	1364 (40.0%)	
Non intrapartum n (%)	127 (45.0%)	2047 (60.0%)	
Anesthesia			<0.001
Spinal n (%)	80 (28.4%)	1730 (50.47%)	
CSE n (%)	70 (24.8%)	785 (23.0%)	
DPE n (%)	4 (1.4%)	49 (1.4%)	
Epidural top-up n (%)	128 (45.4%)	847 (24.8%)	

CSE=Combined spinal epidural; DPE=Dural puncture epidural; SD=Standard Deviation

	Elective			Non elective			Overall		
Anesthesia mode	n/n	% (95%CI)	Median (IQR) pain score	n/n	% (95%CI)	Median (IQR) pain score	n/n	% (95%CI)	Median (IQR) pain score
Spinal	44/1181	3.7 (2.7-5.0)	6 (4-8)	36/629	5.7 (4.0-7.8)	8 (6-9)	80/1810	4.4 (3.5-5.4)	6 (4-9)
Combined Spinal Epidural	42/459	9.2 (6.7-12.2)	6 (4-8)	28/396	7.1 (4.7-10.1)	7 (5-8)	70/855	8.2 (6.4-10.2)	6 (4-8)
Dural puncture epidural	0/3	-	-	4/50	8.0 (2.2-19.2)	7.5 (6.5-9)	4/53	7.5 (2.1-18.2)	7.5 (6.5-9)
Epidural top-up	5/41	12.2 (4.1-26.2)	6 (4.5-7)	123/934	13.2 (11.1-15.5)	7 (4-8.5)	128/975	13.1 (11.1-15.4)	7 (4-8)
Total	91/1684	5.4 (4.3-6.5)	6 (4-8)	191/2009	9.5 (8.2-10.8)	7 (4-9)	282/3693	7.6 (6.8-8.5)	6 (4-8)

Table 2. Incidence of intraoperative pain and numerical pain scores by mode of anesthesia in the elective and non-elective cesarean deliveries.

Incidence $p < 0.0001$. CI Confidence interval. IQR- Interquartile range

Variable	Adjusted		
	Odds ratio	95% CI	p value
Age			
20-29	Ref		
<20	0.64	0.19, 2.16	0.469
30-34	0.90	0.63, 1.26	0.544
35-39	0.82	0.57, 1.17	0.265
>=40	1.19	0.78, 1.84	0.420
Hispanic			
Non-Hispanic	Ref		
Hispanic	0.86	0.52, 1.43	0.558
Missing	0.63	0.24, 1.69	0.361
Racial group			
White	Ref		
Asian	0.90	0.59, 1.38	0.639
Black or African American	1.05	0.69, 1.60	0.803
Other Race	1.10	0.71, 1.71	0.668
Missing	1.05	0.35, 3.12	0.932
Preferred language			
English	Ref		
Spanish	1.79	1.02, 3.14	0.044
Other	0.83	0.46, 1.48	0.521
Missing	1.74	0.75, 4.04	0.195
CD urgency			
Elective	Ref		
Urgent	1.24	0.82, 1.88	0.312
Emergency	1.33	0.79, 2.24	0.276
Intrapartum cesarean	0.92	0.58, 1.46	0.733
Anesthesia			
Spinal	Ref		
Combined spinal epidural	2.01	1.40, 2.89	<0.001
Dural puncture epidural	1.39	0.44, 4.39	0.575
Epidural top-up	2.92	1.93, 4.41	<0.001

Table 3. Adjusted odds ratios of measured variables for intraoperative pain.

CD=cesarean delivery; CI Confidence interval

Figure 1

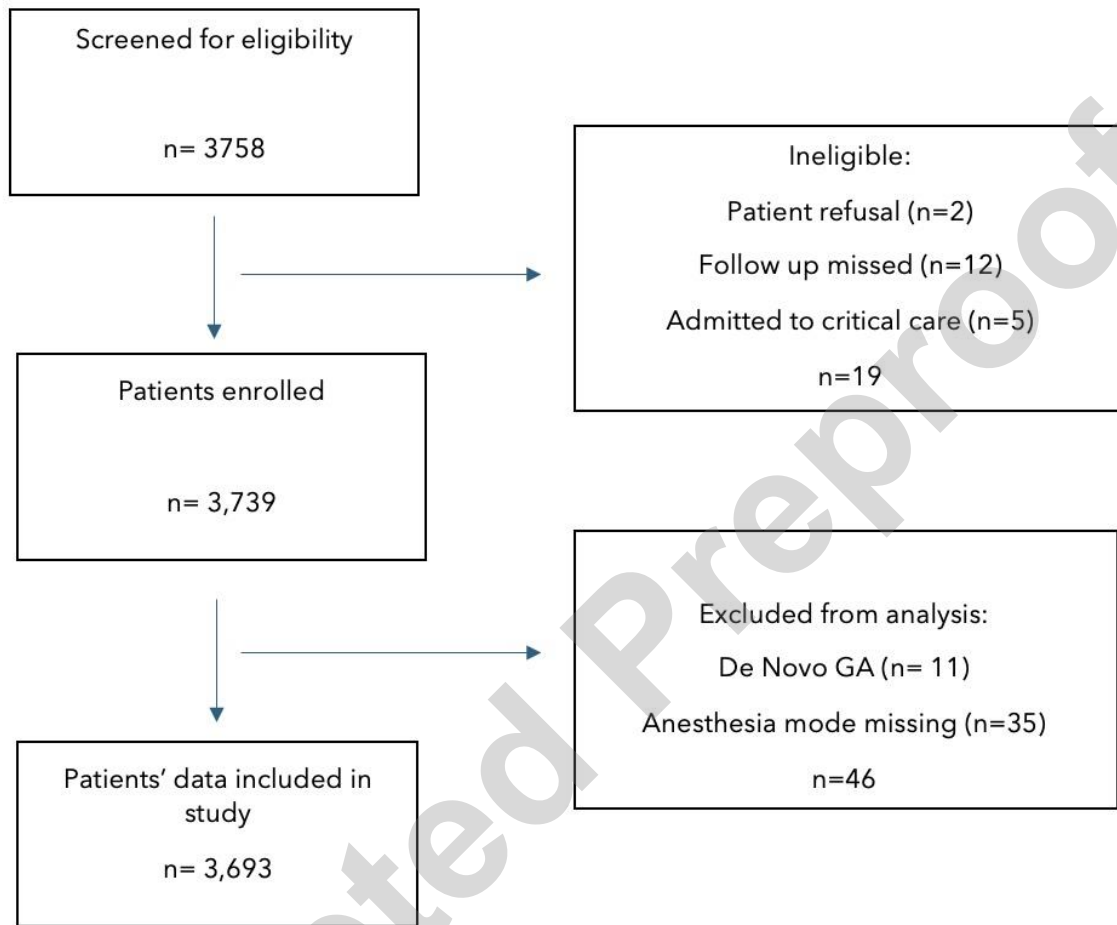


Figure 2a-b

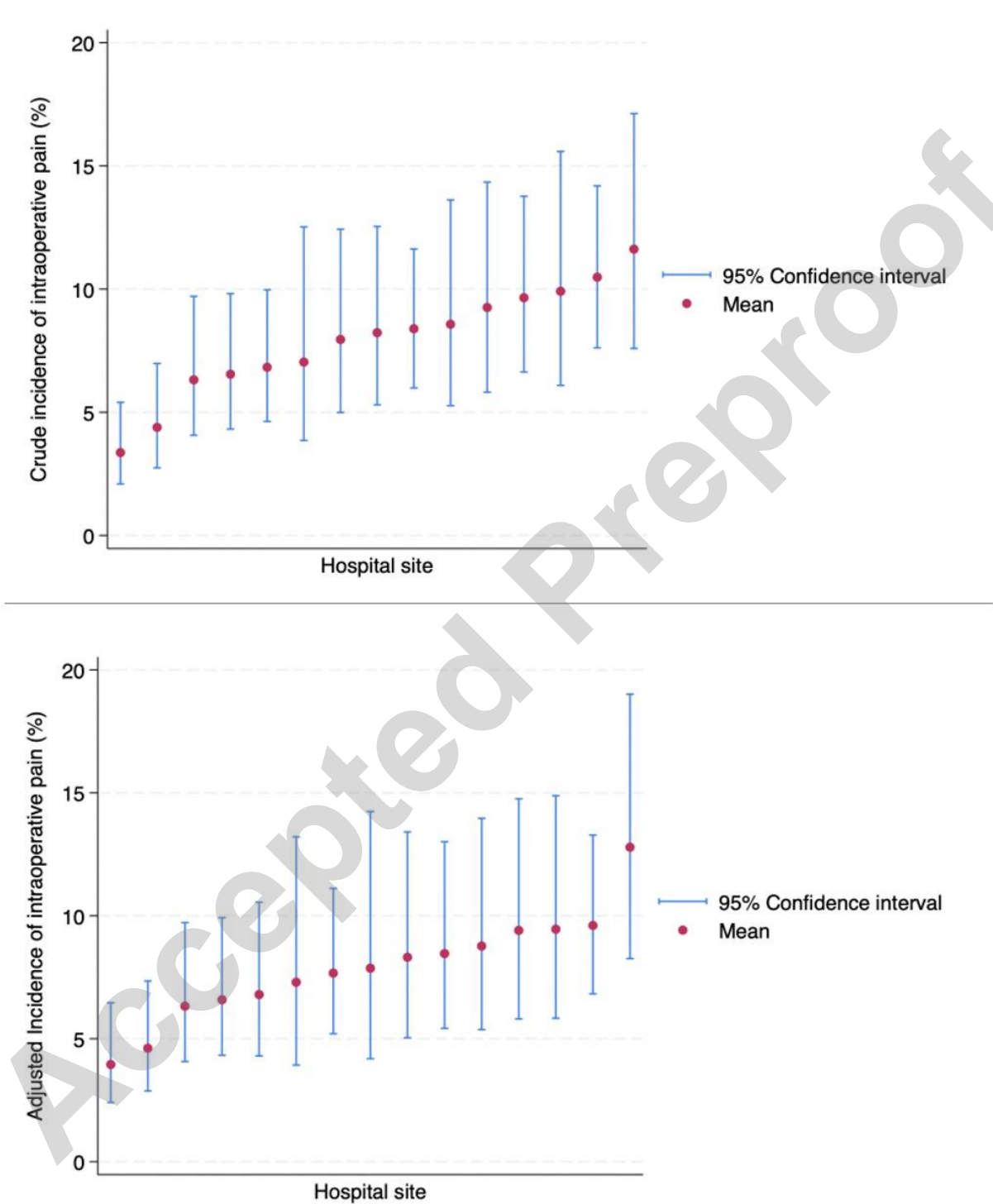
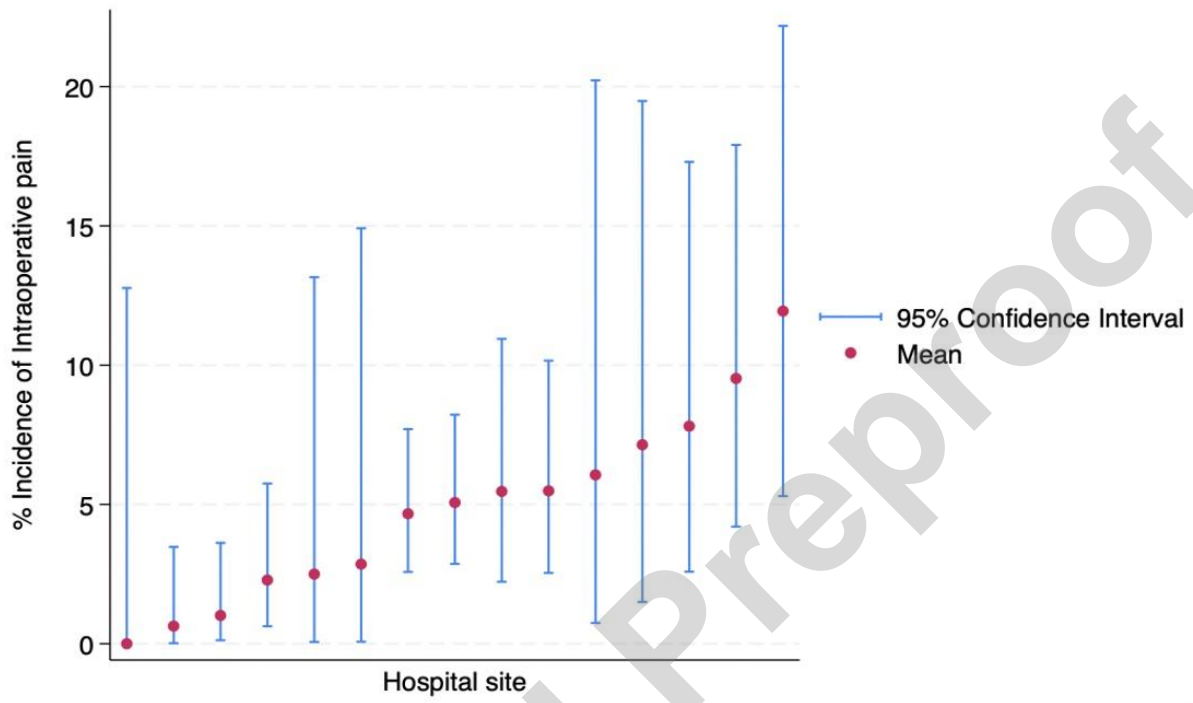


Figure 3



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Collaborators gained local IRB approval, consent for data, collected and verified patient data, uploaded data and/or prepared data transfer and agreements between parent organisations.

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